

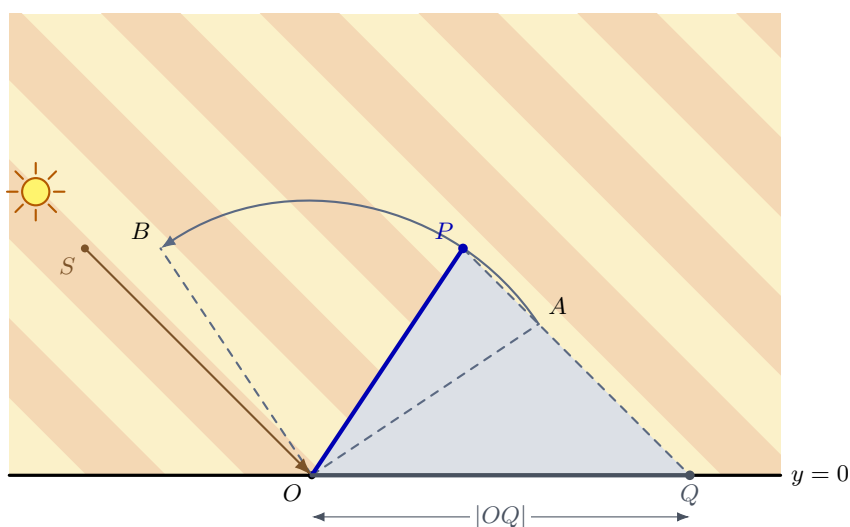
Problem A

Natsuhikage

Time limit: 2 seconds

A team's strength has many dimensions, but a contest ranking sees only one projection. On the contest floor, perhaps some directions remain under your control; yet sometimes the sunlight in Baku simply is not on your side...

A fixed-length rod is hinged to the ground. It rotates between two boundary orientations and otherwise remains above the ground. Sunlight consists of parallel rays traveling in a fixed direction. Find the minimum and maximum shadow lengths over all allowed orientations.



In the figure, O is the origin, $\vec{SO}$ is the direction of sunlight, and the rod may rotate counterclockwise from $\vec{OA}$ to $\vec{OB}$. The current rod is OP , and its shadow length is $|OQ|$.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), the number of test cases. The description of the test cases follows.

Each test case contains six integers s_x, s_y, a_x, a_y, b_x , and b_y , defining $S = (s_x, s_y)$, $A = (a_x, a_y)$, and $B = (b_x, b_y)$ ($-10^5 \leq s_x, a_x, b_x \leq 10^5$; $1 \leq s_y, a_y, b_y \leq 10^5$).

It is guaranteed that $|OA| = |OB|$, and that $\vec{OB}$ is counterclockwise from or coincides with $\vec{OA}$.

Output

For each test case, output two real numbers: the minimum and maximum shadow lengths, in this order.

Your answer is considered correct if its absolute or relative error does not exceed 10^{-6} .

Sample Input 1

```
2
-3 3 3 2 -2 3
0 1 1 1 -1 1
```

Sample Output 1

```
1.0000000000000000 5.099019513592784
0.0000000000000000 1.0000000000000000
```

Problem B

Slot Machine

Time limit: 1 seconds

An arcade has n slot machines, numbered $1, 2, \dots, n$. At the beginning of the game, the initial prize of each machine is an independently and uniformly chosen **real number** from the interval $[0, m]$. Your cheating method lets you see the current prize of every machine before each choice.

Your strategy is to play the machine with the largest current prize: machines with larger prizes are ranked higher, and ties are broken in favor of the machine with the larger index. After paying its prize, the chosen machine independently generates a new prize uniformly at random from $[0, m]$. Your cheating method reveals the new prize immediately, while the prizes of the other machines remain unchanged.

Determine the expected total value of the prizes you will receive in exactly k plays.

Input

The first line contains three integers n , m , and k ($1 \leq n, k \leq 300, 1 \leq m \leq 10^9$).

Output

Output the expected total value of the k prizes. Your answer is considered correct if its absolute or relative error does not exceed 10^{-6} .

Sample Input 1

1 10 3	15.0000000000
--------	---------------

Sample Output 1

Sample Input 2

2 6 2	7.5000000000
-------	--------------

Sample Output 2

In the first sample, there is only one machine. The expected prize on each play is 5, so the expected total is 15. In the second sample, the expected prizes on the first and second plays are 4 and 3.5, respectively, so the expected total is 7.5.

Problem C

Slot Machine 2

Time limit: 4 seconds

The arcade still has n slot machines, numbered $1, 2, \dots, n$. The owner discovered your cheating and changed the machines' program. At the beginning of the game, the initial prize of each machine is now an independently and uniformly chosen **integer** from $0, 1, \dots, m$. However, your cheating method is sophisticated enough that you can still see the current prize of every machine before each choice.

To avoid giving yourself away by always taking the largest prize, you use a smarter strategy: on the i -th play, you choose the machine with the a_i -th largest current prize: machines with larger prizes are ranked higher, and ties are broken in favor of the machine with the larger index. After paying its prize, the chosen machine independently generates a new prize uniformly at random from the integers $0, 1, \dots, m$. Your cheating method reveals the new prize immediately, while the prizes of the other machines remain unchanged.

Determine the expected total value of the prizes you will receive in exactly k plays.

Input

The first line contains three integers n, m , and k ($1 \leq n, k \leq 300, 1 \leq m \leq 10^9$).

The second line contains k integers $a_1, a_2, \dots, a_k$ ($1 \leq a_i \leq n$).

Output

It can be shown that the exact answer can be written as an irreducible fraction $\frac{p}{q}$, where $q \not\equiv 0 \pmod{10^9 + 7}$. Output the integer x ($0 \leq x < 10^9 + 7$) such that $x \cdot q \equiv p \pmod{10^9 + 7}$.

Sample Input 1

1 10 3
1 1 1

Sample Output 1

15

Sample Input 2

2 2 2
1 2

Sample Output 2

777777785

In the first sample, there is only one machine. The expected prize on each play is 5, so the expected total is 15. In the second sample, the expected total is $\frac{16}{9}$. Since $9^{-1} \equiv 111\,111\,112 \pmod{10^9 + 7}$, the required output is $16 \cdot 111\,111\,112 \bmod (10^9 + 7) = 777\,777\,785$.

Problem D

Strict Majority

Time limit: 5 seconds

A park has n trail junctions connected by trails. Between any two junctions, there is exactly one route that does not visit a junction more than once.

Each junction will have a sign bearing one lowercase English letter. Some signs have already been painted; the missing letters are represented by question marks. You must replace every question mark with a lowercase English letter without changing any letter already in place.

Consider any route passing through at least three junctions. A letter has a strict majority on this route if it appears at more than half of its junctions. If one letter has a strict majority on such a route, visitors will be confused, so this is not allowed.

Complete the signs, or determine that this is impossible.

Input

The first line contains an integer T ($1 \leq T \leq 1000$), the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n ($3 \leq n \leq 5000$), the number of junctions.

The second line of each test case contains a string s of length n . The i -th character of s is either a lowercase English letter already painted on the sign at junction i , or $?$ if its letter has not been painted yet.

The i -th of the next $n - 1$ lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n$), denoting a trail between junctions u_i and v_i .

It is guaranteed that the given edges of each test case form a tree.

It is guaranteed that the sum of n over all test cases does not exceed 5000.

Output

For each test case, output a valid labeling as follows.

If there is no solution, output -1 .

Otherwise, output a string of length n . Its i -th character must be the letter painted on the sign at junction i . If there are multiple solutions, output any of them.

Sample Input 1

Sample Output 1

2	abcd
5	-1
a?c??	
1 2	
2 3	
2 4	
4 5	
5	
aab?a	
1 2	
2 3	
3 4	
4 5	

Problem E

Splendor

Time limit: 2 seconds

In the board game Splendor, players take gems of different colors as they build their trading empires. Its original rules are rather involved, so this problem uses only the simplified rules described below. No other rule from the board game applies.

There are five colors of gems, with an unlimited supply of each color. Your goal is to collect exactly a_i gems of color i .

On each turn, you must perform exactly one of the following actions:

- take one gem of each of three distinct colors;
- take two gems of the same color.

You may never take more than the required number of gems of any color. Find the minimum number of turns needed to reach your goal, or determine that it is impossible.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), the number of test cases. The description of the test cases follows.

Each test case contains five integers $a_1, a_2, \dots, a_5$ ($0 \leq a_i \leq 10^9$), where a_i is the required number of gems of color i .

Output

For each test case, output an integer: the minimum number of turns needed to collect exactly the required gems. If there is no solution, output -1 .

Sample Input 1

3	2
2 2 2 0 0	2
3 1 1 0 0	0
0 0 0 0 0	

Sample Output 1

Sample Input 2

2	-1
1 0 0 0 0	2
2 1 1 1 1	

Sample Output 2

In the first test case of the first sample, two turns can each take one gem of colors 1, 2, and 3.

In the second test case of the first sample, one turn can take one gem of colors 1, 2, and 3, and another turn can take two gems of color 1.

In the first test case of the second sample, a single gem cannot be collected by either allowed action.

In the second test case of the second sample, the two turns can take colors 1, 2, and 3, then colors 1, 4, and 5.

Problem F

Occurrences

Time limit: 4 seconds

You are building an index for a manuscript represented by a lowercase string S of length n . The characters of S are indexed from 1 to n .

For any nonempty string T , let $\text{occ}(T)$ be the number of starting positions p with $1 \leq p \leq n - |T| + 1$ such that

$$S[p \dots p + |T| - 1] = T.$$

Occurrences may overlap.

Each query specifies an interval $[l, r]$. Consider every nonempty substring $S[i \dots j]$ whose endpoints satisfy $l \leq i \leq j \leq r$. The answer to the query is

$$\sum_{l \leq i \leq j \leq r} \text{occ}(S[i \dots j]).$$

Input

The first line contains an integer n ($1 \leq n \leq 2 \times 10^5$).

The second line contains the string S of length n , consisting only of lowercase English letters.

The third line contains an integer q ($1 \leq q \leq 2 \times 10^5$).

Each of the next q lines contains two integers l and r ($1 \leq l \leq r \leq n$).

Output

For each query, output an integer, the sum defined above.

Sample Input 1

```
4
abab
3
1 2
1 4
2 3
```

Sample Output 1

```
6
16
5
```

Sample Input 2

```
4
aaaa
3
1 1
1 2
2 4
```

Sample Output 2

```
4
11
20
```

In the first query of the first sample, the three substrings in the sum are $S[1 \dots 1] = a$, $S[2 \dots 2] = b$, and $S[1 \dots 2] = ab$. Their occurrence counts in S are 2, 2, and 2, respectively, so the answer is 6.

In the second query of the second sample, both $S[1 \dots 1]$ and $S[2 \dots 2]$ equal a , and each contributes $\text{occ}(a) = 4$. Also, $S[1 \dots 2] = aa$ contributes $\text{occ}(aa) = 3$. Therefore, the answer is $4 + 4 + 3 = 11$.

Problem G

Yet Another A Plus B Problem

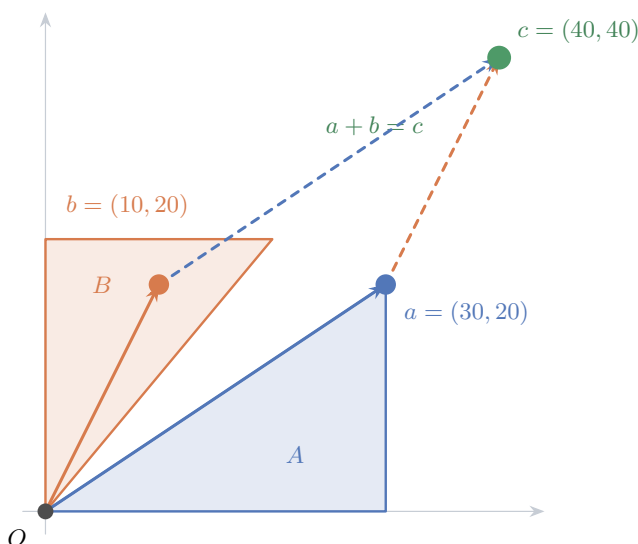
Time limit: 10 seconds

You are given two possibly degenerate polygons A and B . Each polygon includes its boundary and interior. A degenerate polygon may be a segment or a point.

For two points $a = (x_a, y_a)$ and $b = (x_b, y_b)$, define their sum as

$$a + b = (x_a + x_b, y_a + y_b).$$

For each query point c_i , determine whether there are points $a \in A$ and $b \in B$ such that $a + b = c_i$. If they exist, provide one such pair.



The figure illustrates the first test case of the first sample.

Input

The first line contains an integer T ($1 \leq T \leq 200$), the number of test cases. The description of the test cases follows.

The first line of each test case contains three integers n, m , and q ($1 \leq n, m, q \leq 200$), the number of vertices of A , the number of vertices of B , and the number of queries, respectively.

The second line contains $2n$ integers $x_{a_1}, y_{a_1}, x_{a_2}, y_{a_2}, \dots, x_{a_n}, y_{a_n}$, describing the vertices of A in counterclockwise order.

The third line contains $2m$ integers $x_{b_1}, y_{b_1}, x_{b_2}, y_{b_2}, \dots, x_{b_m}, y_{b_m}$, describing the vertices of B in counterclockwise order.

The i -th of the next q lines contains two integers x_{c_i} and y_{c_i} , the coordinates of the query point c_i .

All input coordinates are between 0 and 2×10^6 , inclusive. Within each polygon, all given vertices are distinct. A polygon with one vertex represents that point, and a polygon with two vertices represents the segment joining them. A polygon with at least three vertices is simple, has positive area and a connected interior, and has no three cyclically consecutive collinear vertices.

It is guaranteed that $\sum n \leq 500$ and $\sum m \leq 500$ over all test cases. Please note that there is no additional constraint on the sum of q over all test cases.

Output

For each query, output one line.

If a solution exists, output four real numbers x_a , y_a , x_b , and y_b , representing points $a = (x_a, y_a) \in A$ and $b = (x_b, y_b) \in B$ such that $a + b = c_i$. Otherwise, output $-1 \ -1 \ -1 \ -1$.

An output pair is accepted if the Euclidean distance from each output point to its corresponding polygon does not exceed 10^{-6} and the Euclidean distance from $a + b$ to c_i does not exceed 10^{-6} .

Sample Input 1

```

4
3 3 3
0 0 30 0 30 20
0 0 20 24 0 24
40 40
31 0
31 2
2 2 3
0 0 0 300
0 0 300 0
1 2
2 1
301 0
3 4 2
0 0 1 0 0 1
0 0 1 0 1 1 0 1
2 1
1 2
8 3 3
0 0 12 6 3 6 7 4 1 1 1 5 2 6 0 6
0 0 2 0 0 1
2 7
5 5
4 5
  
```

Sample Output 1

```

30 20 10 20
-1 -1 -1 -1
29.5 0 1.5 2
0 2 1 0
0 1 2 0
-1 -1 -1 -1
1 0 1 1
0 1 1 1
2 6 0 1
5 5 0 0
-1 -1 -1 -1
  
```

Sample Input 2

```

1
1 1 1
2 3
4 5
6 8
  
```

Sample Output 2

```

2 3 4 5
  
```

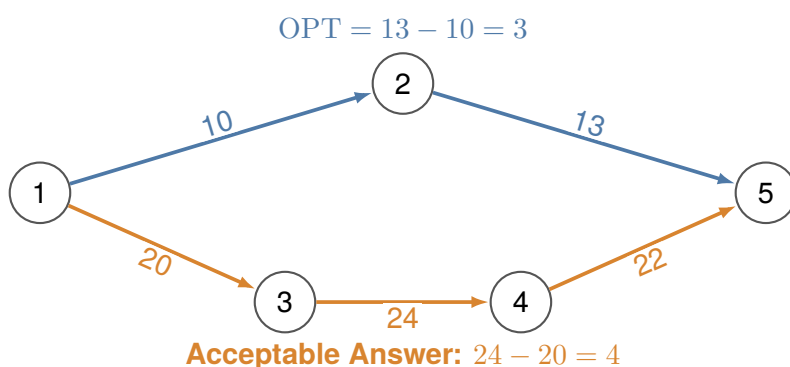

Problem H

Closing the Gap... Approximately

Time limit: 4 seconds

Close Enough Logistics is planning a route for its trucks. Its transportation network consists of junctions and one-way roads, and each road has an integer speed limit. To avoid repeatedly speeding up and slowing down, the company wants the speed limits along the route to be as close as possible. The gap of a path is its highest speed limit minus its lowest.

Given two junctions, s and t , let OPT be the minimum gap of any path from s to t . In the spirit of “close enough is good enough,” you do not have to find an optimal path. A $\sqrt{2}$ -approximation will do: find a path from s to t whose gap is at most $\sqrt{2} \cdot OPT$.



First sample: the blue path has gap $13 - 10 = 3 = OPT$.
 The orange path has gap $24 - 20 = 4$ and is acceptable because $4 \leq \sqrt{2} \cdot 3$.

Input

The first line contains four integers n, m, s , and t ($2 \leq n \leq 10^5, 1 \leq m \leq 2 \times 10^5, 1 \leq s, t \leq n, s \neq t$), denoting the number of junctions, the number of roads, the starting junction, and the destination, respectively.

The i -th of the next m lines contains three integers u_i, v_i , and w_i ($1 \leq u_i, v_i \leq n, 1 \leq w_i \leq 10^9$), denoting a one-way road from junction u_i to junction v_i with speed limit w_i .

It is guaranteed that the graph has no self-loops or multiple edges, and that there is a path from s to t .

Output

On the first line, output an integer k ($2 \leq k \leq n$), the number of vertices on your path.

On the second line, output k distinct integers $p_1, p_2, \dots, p_k$ ($1 \leq p_i \leq n$), the vertices of the path in order. They must satisfy $p_1 = s, p_k = t$, and there must be a directed edge from p_i to p_{i+1} for every $1 \leq i < k$.

Let ANS be the gap of the output path. Your answer is accepted if $ANS^2 \leq 2 \cdot OPT^2$. If there are multiple solutions, output any of them.

Sample Input 1

```
5 5 1 5
1 2 10
2 5 13
1 3 20
3 4 24
4 5 22
```

Sample Output 1

```
3
1 2 5
```

Sample Input 2

```
4 5 1 4
1 2 3
2 4 8
1 3 5
3 4 7
2 3 6
```

Sample Output 2

```
3
1 3 4
```

Problem I

Choose a Name

Time limit: 5 seconds

Pigstd recently vibe-coded a user management system. The database is, in fact, a file stored directly on disk. Its username lookup treats the query as a prefix: searching for a string returns every user whose username begins with that string.

Consequently, if one username is a prefix of another, an exact lookup for the shorter username also finds the longer one. Pigstd remembers prefix-free codes and decides to impose the same requirement on usernames. A set of usernames is prefix-free if, for every two different users, neither username is a prefix of the other. A string is considered a prefix of itself, so two users cannot have equal usernames.

There are n existing users. The i -th user provides two candidate usernames: a left candidate l_i and a right candidate r_i . Choose exactly one candidate for every user so that all chosen usernames form a prefix-free set, or determine that this is impossible.

Input

The first line contains an integer T ($1 \leq T \leq 10^4$), the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n ($1 \leq n \leq 5 \times 10^5$).

The i -th of the next n lines contains two nonempty strings l_i and r_i , the left and right candidates of the i -th user. Each string consists only of lowercase English letters.

It is guaranteed that the sum of n over all test cases does not exceed 5×10^5 .

It is guaranteed that the sum of the lengths of all strings over all test cases does not exceed 10^6 .

Output

For each test case, output YES if the answer exists, and NO otherwise.

If the answer exists, on the next line, output a string s of length n consisting only of the characters `l` and `r`. If s_i is `l`, the i -th user receives l_i ; if s_i is `r`, the i -th user receives r_i .

If there are multiple solutions, output any of them.

Sample Input 1

Sample Output 1

3	YES
3	lll
alice al	NO
bob bobby	YES
carol cat	lrr
2	
a ab	
a ab	
3	
a dog	
ab cat	
abc cow	

Sample Input 2**Sample Output 2**

2	YES
1	l
x x	YES
4	lrrr
a za	
ab zb	
abc zc	
abcd zd	

In the first test case of the first sample, the chosen usernames are `alice`, `bob`, and `carol`. None is a prefix of another. In the second test case, every possible pair of chosen usernames is either equal or consists of `a` and `ab`, so no valid choice exists.

In the second sample, a test case with one user is always valid. In its second test case, choosing `za`, `zb`, `zc`, and `zd` produces a prefix-free set.

Problem J

Joyride

Time limit: 2 seconds

You are planning joyrides through a city with n intersections connected by m two-way roads.

A joyride starts at an intersection, follows at least 3 roads, and returns to its starting intersection. It does not use any road more than once or visit any other intersection more than once. Thus, every joyride follows a simple cycle.

For a fixed intersection i , choose as many joyrides through i as possible, subject to no road belonging to more than one chosen joyride. For each $i = 1, 2, \dots, n$, determine the maximum number of joyrides that can be chosen.

Input

The first line contains an integer T ($1 \leq T \leq 2 \times 10^5$), the number of test cases. The description of the test cases follows.

The first line of each test case contains two integers n and m ($1 \leq n \leq 2 \times 10^5, 0 \leq m \leq 2 \times 10^5$), the number of intersections and roads, respectively.

Each of the next m lines contains two integers u and v ($1 \leq u, v \leq n, u \neq v$), describing a two-way road between intersections u and v .

Intersections are numbered from 1 to n . No road connects an intersection to itself, and there is at most one road between any pair of intersections. The graph is not necessarily connected.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 . It is guaranteed that the sum of m over all test cases does not exceed 2×10^5 .

Output

For each test case, output n integers separated by a space. The i -th integer is the maximum number of joyrides through intersection i that can be chosen with no road belonging to more than one of them.

Sample Input 1

```
3
5 6
1 2
2 3
3 1
1 4
4 5
5 1
5 6
1 2
2 5
1 3
3 5
1 4
4 5
4 2
1 2
3 4
```

Sample Output 1

```
2 1 1 1 1
1 1 1 1 1
0 0 0 0
```

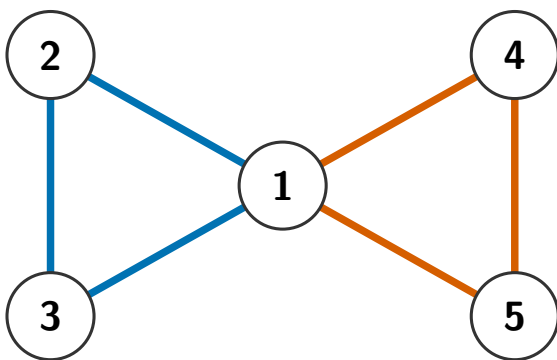
In the first test case, we can choose 2 edge-disjoint joyrides through intersection 1: $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ and $1 \rightarrow 4 \rightarrow 5 \rightarrow 1$. On the other hand, every joyride through intersection 1 uses two roads incident to it, and

intersection 1 has only four incident roads. Therefore, no more than two such joyrides can be chosen, so the answer for intersection 1 is 2.

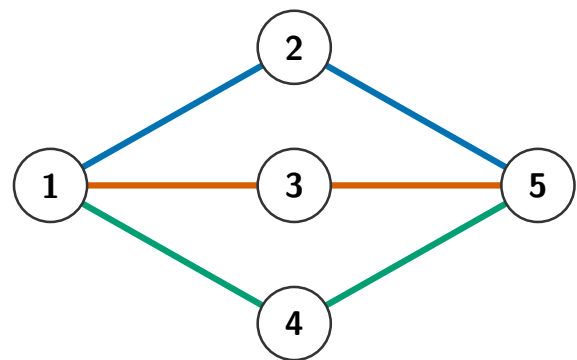
Each intersection from 2 through 5 lies on one of the two joyrides above, so its answer is at least 1. Each of these intersections has only two incident roads, both of which are used by any joyride through it, so a second edge-disjoint joyride cannot be chosen. Therefore, their answers are all 1.

In the second test case, there are three edge-disjoint paths from intersection 1 to intersection 5, passing through intersections 2, 3, and 4, respectively. Every joyride through intersection 1 must combine two of these paths into a cycle. Any two such cycles share at least one path, so they cannot both be chosen. Thus, the answer for intersection 1 is 1.

Every intersection lies on at least one cycle, but no intersection can belong to two edge-disjoint joyrides. Therefore, all five answers are 1.



The two edge-disjoint routes in the first test case.



The three paths from intersection 1 to intersection 5 in the second test case.

In the first diagram, the blue route is $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$, and the orange route is $1 \rightarrow 4 \rightarrow 5 \rightarrow 1$. They share only intersection 1 and do not share any road. In the second diagram, the three colors show the three paths from intersection 1 to intersection 5.

Problem K

Team Formation

Time limit: 2 seconds

At the start of the new semester, $3n$ first-year students attend tryouts for the university's competitive programming team. They hope to compete in ICPC/CCPC, but first they must form teams. The teams must satisfy two rules:

- every team contains exactly three students;
- every student belongs to exactly one team.

The rules make it easy to form teams, but not necessarily the right teams. Every pair of students has an integer compatibility score, which may be negative when the two would rather not work together. A team's score is the sum of the three pairwise compatibility scores among its members.

Find the maximum possible total score over all valid team assignments.

Input

The first line contains an integer n ($1 \leq n \leq 8$).

The i -th of the next $3n$ lines contains $3n$ integers $a_{i,1}, a_{i,2}, \dots, a_{i,3n}$ ($-10^8 \leq a_{i,j} \leq 10^8$). The integer $a_{i,j}$ is the compatibility score between students i and j .

It is guaranteed that $a_{i,i} = 0$ and $a_{i,j} = a_{j,i}$ for all $1 \leq i, j \leq 3n$.

Output

Output the maximum possible sum of scores of all teams.

Sample Input 1

2 0 5 4 -1 -2 -3 5 0 3 -2 -2 -3 4 3 0 -1 -4 -2 -1 -2 -1 0 7 6 -2 -2 -4 7 0 5 -3 -3 -2 6 5 0	30
---	----

Sample Output 1

Sample Input 2

1 0 -5 2 -5 0 1 2 1 0	-2
--------------------------------	----

Sample Output 2

In the first sample, one optimal team assignment consists of teams $(1, 2, 3)$ and $(4, 5, 6)$. Their total score is $(5 + 4 + 3) + (7 + 6 + 5) = 30$.

In the second sample, all three students must form the only team. Its score is $-5 + 2 + 1 = -2$.

Problem L

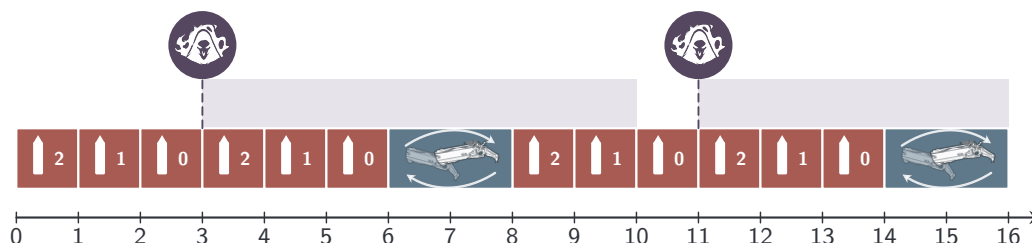
Reaper

Time limit: 2 seconds

Reaper wants to know how quickly he can fire his *Hellfire Shotguns* if the fight lasts forever.

The weapon holds m rounds when fully loaded. Firing one round consumes one round and takes exactly one second. *Reaper* may reload between shots by discarding the weapon he is holding and drawing a new one loaded with m rounds. This takes r seconds.

Reaper can also activate *Wraith Form*, but not while he is firing or reloading. Activating it takes no time and immediately restores the weapon to m rounds. *Wraith Form* is available at time zero. After each activation, it enters a c -second cooldown and cannot be activated during that time. Its cooldown continues to elapse while *Reaper* is firing, reloading, or waiting.



The first test case in the sample is illustrated above. It can be shown that the optimal pattern repeats between consecutive reloads, firing six rounds every eight seconds, so the answer is $6/8 = 0.75$.

Reaper may choose when to fire, reload, activate *Wraith Form*, or wait. Let $f(n)$ be the maximum number of rounds *Reaper* can finish firing during the first n seconds, starting with the weapon fully loaded and *Wraith Form* available. It can be shown that the following limit exists:

$$\lim_{n \rightarrow \infty} \frac{f(n)}{n}.$$

Find this limit, which is *Reaper*'s maximum effective firing rate in rounds per second.

Input

The first line contains an integer T ($1 \leq T \leq 100$), the number of test cases. The description of the test cases follows.

Each test case contains three integers m , r , and c ($1 \leq m, r, c \leq 100$): the number of rounds the weapon holds when fully loaded, the duration of a reload in seconds, and the cooldown of *Wraith Form* in seconds, respectively.

Output

For each test case, output the maximum effective firing rate.

Your answer is considered correct if its absolute or relative error does not exceed 10^{-9} .

Sample Input 1

Sample Output 1

5	0.750000000000
3 2 7	0.666666666667
1 1 2	0.666666666667
2 100 3	0.250000000000
1 100 4	0.666666666667
4 100 6	

Problem M

Brackets

Time limit: 2 seconds

You are given two regular bracket sequences S and T of length n , and two arrays a and b .

A bracket sequence is called regular if it can be transformed into a valid arithmetic expression by inserting 1 and $+$ between its characters.

The weight of a bracket sequence P of length n is defined as follows:

- for every position i such that $P_i = \text{[(]}$, add a_i ;
- for every position i such that $P_i = \text{[)]}$, add b_i .

In other words,

$$\text{weight}(P) = \sum_{P_i = \text{[(]}} a_i + \sum_{P_i = \text{[)]}} b_i.$$

For each integer k from 0 to n , choose exactly k positions and replace S_i with T_i at every chosen position i .

Let the resulting sequence be S' . Among all choices of exactly k positions such that S' is still a regular bracket sequence, find the maximum possible weight of S' .

If it is impossible to obtain a regular bracket sequence for some k , output -1 for this k .

Input

The first line contains an integer T ($1 \leq T \leq 250\,000$), the number of test cases. The description of the test cases follows.

Each test case consists of five lines.

The first line of each test case contains an even integer n ($2 \leq n \leq 5 \times 10^5$), the length of the bracket sequences.

The second line contains a regular bracket sequence S of length n , consisting only of [(] and [)] .

The third line contains a regular bracket sequence T of length n , consisting only of [(] and [)] .

The fourth line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

The fifth line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed 5×10^5 .

Output

For each test case, output $n + 1$ integers separated by a space.

The $(k + 1)$ -th integer, for $0 \leq k \leq n$, should be the maximum possible weight after choosing exactly k positions and replacing S_i with T_i at all chosen positions.

If no regular bracket sequence can be obtained for a given k , the $(k + 1)$ -th integer must be -1 .

Sample Input 1

```

5
2
()
()
1 2
3 4
6
((()))
()()()
1 2 3 4 5 6
6 5 4 3 2 1
4
()()
(())
1 5 4 6
2 2 5 5
8
(())()()
()((()()))
8 5 1 4 2 6 7 3
1 9 7 6 6 5 9 2
10
((()())())
(((())(())))
2 3 4 1 9 5 8 7 1 10
1 9 6 7 6 2 8 5 3 10
  
```

Sample Output 1

```

5 5 5
12 12 18 18 18 18 18
12 12 16 16 16
42 42 45 45 47 47 47 45 45
48 48 52 52 53 53 53 53 53 53
  
```

In the first test case, replacing characters at any chosen positions does not change the sequence. Its weight is $a_1 + b_2 = 1 + 4 = 5$.

In the second test case, only positions 2 and 5 differ between S and T . Replacing either one alone makes the sequence irregular, while replacing both changes the sequence into T and increases its weight from 12 to 18. The other four positions can be chosen without changing the sequence.